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LETTER

A more complete IMAGE: enhancing the climate change impact representation in integrated assessment models

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Abstract

Currently, climate impacts are mostly absent in the analysis of process-based integrated assessment models (IAMs). This is becoming increasingly problematic given the fact that climate impacts are expected to influence both baseline developments as well as the capacity to mitigate and adapt in the future. This paper presents a set of scenarios made using the IAM IMAGE with climate change impacts on labor productivity and GDP, renewable energy supply, heating and cooling demand, food production, water availability, and biodiversity. The results show that these impacts can have significant effect on existing projections of the economic, energy and land system that are usually not included in IAM projections. In the current implementation, the aggregated economic impacts are often larger in size than sectoral feedbacks, such as those on energy supply and demand, GHG emissions, and food consumption. Importantly, we find that there are several hotspot areas, including Western Africa, Indonesia and the Middle East, where multiple risks accumulate more than in other regions.

1. Introduction

Climate change represents a significant threat to human well-being and sustainable development [1]. Mitigation and adaptation form two key response options to this threat. Mitigation is aimed at limiting climate change by reducing emissions or enhancing the sinks of greenhouse gases. Adaptation is defined by actions to adjust to the actual climate conditions, to reduce harm or exploit new opportunities. There are connections between these strategies, with potential synergies and trade-offs, while also climate change itself can impact both strategies. These linkages thus need to be accounted for in transformation decisions, for instance, in the design of cities, agriculture, and energy systems [2–7]. At a broader level, the absence of effective mitigation measures could render adaptation strategies inadequate and result in residual damages [8]. Similarly, the potential for mitigation depends on the level of climate change, via changes in, for example, energy supply and demand, biomass supply and the potential for reforestation. While some measures might lead to trade-offs between adaptation and mitigation (like air conditioning), others can result in synergy (like afforestation). Therefore, it is important to assess the consequences of mitigation and adaptation decisions and climate impacts in an integrated way. However, current research mostly focuses on specific research domains. The climate system is studied using physical climate models (assessed in IPCC WGI). Impacts and possible adaptation strategies are studied by the so-called 'Vulnerability, Impacts, and Adaptation' community (assessed in IPCC WG II). Drivers of climate change and mitigation strategies are mostly studied in mitigation studies, including integrated assessment model (IAM) (assessed in IPCC WG III).

Community scenarios, such as the shared socio-economic pathways (SSPs), serve as a means to connect these different research communities. In fact, the SSP framework was designed to allow for a comparison of mitigation and adaptation strategies [9, 10]. Still, only a limited portion of the literature focuses on integrated, comprehensive assessment [11, 12]. Some IAMs, known as cost-benefit models, form an exception as they compare the costs of climate impacts and mitigation and sometimes even adaptation (e.g. DICE, FUND and WITCH) [13–15]. These models typically use rather abstract monetary damage curves. Other IAMs, with considerably more process-based detail on activities, typically focus only on the effectiveness of mitigation action. Also, here some exceptions exist, including the integration of impacts into computable general equilibrium models (CGEs) [16, 17], focusing mostly on monetary impacts, or studies on specific topics [18–20]. Interestingly, the elaboration of the SSP scenarios explicitly mentions that climate impacts on human systems have been excluded, mostly for methodological reasons [21]. The models used would also only partly be able to deal with climate impacts. However, interest is growing in scenarios that describe possible futures while accounting for climate impacts, adaptation and mitigation at the same time, as climate change is a relevant factor in the future. In this context, also O'Neill *et al* identify the development of more integrated scenarios as a key priority. Also Monier *et al* [22], Piontek *et al* [23] and van Maanen *et al* [7] argued to create such integrated scenarios and presented some examples on how to do so.

This paper explores how climate impacts can be included in IMAGE 3.2 [24, 25]. The representation of these impacts is, as much as possible, in terms of physical impacts on human activities. This study assesses several key sectoral and regional biophysical climate change impacts as a function of mitigation action and socio-economic developments. Adaptation is only partly included as it requires more extensive model development [7, 26]. The model runs are done for a baseline scenario and scenarios aiming to stay well-below 2 °C. The main purpose is to explore how including climate impacts changes the projected climate mitigation pathways, and to assess which regions are likely to be most affected.

2. Methods

We use the IMAGE model to develop scenarios that combine projections for sectoral activities in scenarios with and without climate change with, multi-sectoral, geographically resolved representations of the biophysical impacts. The climate impact data is partly based on the inter-sectoral impact model intercomparison project (ISIMIP) [27]. The IMAGE projections are based on the SSP2 scenario that follows a medium socio-economic development path [28]. The scenario assessment allows us to estimate how climate change could impact specific sectors and regions. Below we briefly describe the IMAGE 3.2 model and the way climate impacts are included (see figure 1). The supplementary information (S1) provides a more detailed description.

2.1. IMAGE 3.2 and climate impacts

IMAGE is an IAMs framework that simulates the environmental consequences of changes in human activities at the global and regional level. A detailed model documentation is available online (www.pbl.nl/IMAGE). Here, we provide a brief description [29]. The model is designed to analyze large-scale and long-term interactions between human development and the natural environment and to identify response strategies to global environmental change. It consists of a combination of models, with some of them operating in a hard-coupled mode and others connected through soft links (the latter means that models run independently with information exchange via data files). The framework is structured around the causal chain of key global sustainability issues. It consists of two main systems: (1) the human or socio-economic system, describing the long-term development of human activities like transport, agriculture and energy consumption and production, and (2) the earth system, which represents the dynamical behavior of natural systems, such as natural vegetation, carbon, nutrient and water cycles and climatic effects. The two systems are linked through emissions, land use, climate feedback, and potential human policy responses. In terms of resolution, the model simulates most socio-economic parameters for 26 world regions and most environmental parameters based on a geographical grid of 30 by 30 min or 5 by 5 min.

Key inputs into the model include exogenous assumptions on population, economic development, lifestyle, policies, and technological change. These assumptions are used to describe energy production and consumption in the TIMER model and agriculture in th MAGNET model. TIMER is a system-dynamics energy system simulation model. The model describes key trends in energy use and supply. MAGNET is a CGE agriculture system model [30], which uses information from IMAGE on land availability, suitability, and changes in crop yields due to agricultural expansion and climate change. The results from MAGNET on production of crops and animal products and technology-driven yield changes are fed back to IMAGE to calculate spatially explicit land-use change and the environmental impacts on

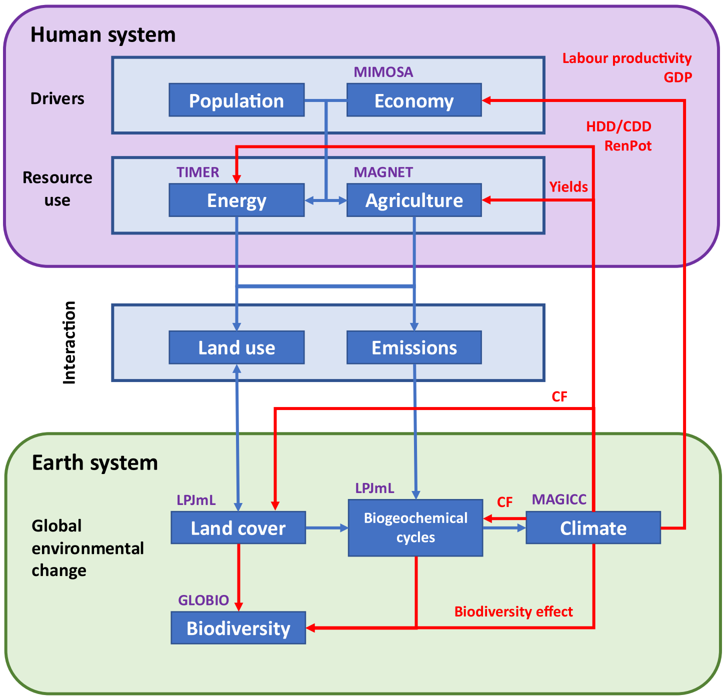

Figure 1. Representation of the IMAGE framework and all climate feedback included in the model. The blue arrows indicate model linkages. Climate feedbacks are indicated by red arrows. The impact on biogeochemical cycles includes temperature and precipitation feedback on the water cycle, atmospheric chemistry and natural emissions. The feedback on land cover includes, directly and indirectly, the feedback on vegetation growth. 'HDD/CDD' stands for heating and cooling degree feedback on cooling and heating demand. 'RenPot' denotes the impacts on the potential and costs of hydropower, bioenergy, wind power and solar power. 'Labour prod.' indicates impacts on labour productivity. 'CF' stands for climate feedback (including CO_2 fertilization). A more detailed description can be found in the supplementary information (S1).

carbon, nutrient and hydrological cycles, biodiversity, and climate.

Emissions of greenhouse gases and air pollutants as well as land-use change form the main interactions between the earth and human system. The LPJmL model [31] forms a key model part of the IMAGE earth system representation. It describes land cover, the terrestrial carbon cycle and vegetation dynamics. The model determines the productivity of natural and cultivated ecosystems at the grid cell level taking into account the variety in plant and crop functional types. The land cover in IMAGE is determined by the output of LPJmL combined with required agricultural production and a set of allocation rules. These rules defined the 'attractiveness' of each cell, based on crop productivity, proximity to other agricultural area, water and infrastructure). Subsequently, both the energy, agricultural production and land use data can be used to calculate emissions of greenhouse gases and air pollutants. Emissions form the basis to calculate the concentrations of greenhouse gases, ozone precursors and species involved in aerosol formation on a global scale. The reduced-complexity climate model MAGICC 6.3 is used to calculate global mean temperature change [32]. Finally, using a pattern-scaling approach, the changes in temperature and precipitation in each grid cell can be derived from the global mean temperature. This approach uses the outcomes of detailed Earth system models and applies a linear scaling algorithm [33]. The model accounts for several feedback mechanisms between climate change and energy, land and vegetation dynamics.

Table 1 lists the climate impacts that are included in this analysis. First, spatially explicit climate-change patterns directly impact the IMAGE-land and coupled LPJmL model on a gridded level, where it affects water demand and availability, crop yields, bioenergy potential and the carbon cycle. Second, the projected climate impacts on GDP and labor productivity are introduced via the MIMOSA model [34], based on a long-term Ramsey-type growth model. The results from Dasgupta *et al* were used for climate impacts on labor productivity in the production function. Here, we use a differentiation between sheltered and non-sheltered jobs given the possibilities for adaptation in the former (accounting for structural change in the economic projections). The remaining impacts on economic growth are accounted for by combining the global impact data from the global damage function of Howard and Sterner [35] with the regional and sectoral contribution of Bosello and Decian [36]. The combination allowed us to scale the regional and sectoral data to Howard and Sterner's global damages and remove the contributions from labor productivity and agriculture to avoid double counting. The resulting GDP trajectories were

Table 1. Included climate impact indicators and their underlying approaches/data (a more detailed description can be found in the supplementary information).

Indicator	Approach
Labour productivity	Changes in performance during working hours [42] translated into changes in GDP
Other economic impacts	GDP loss. Based on the meta-study-based global damage function [35], scaled to the regional level using the data from Bosello and Decian [36].
Renewable supply (wind, PV, CSP, hydro, bioenergy)	Different costs supply curves based on 0.5 × 0.5 grid calculations [37]
Heating/cooling demand	Impact via population-weighted HDD/CDD, based on 0.5 × 0.5 grid [38, 39]
Crop yields	Crop yield change due to climate change is calculated in LPJmL on a 0.5 × 0.5 grid [31] and run through the MAGNET model to determine changes in agriculture markets and food consumption and GDP [43].
Water stress	Water stress based on water availability and water use from LPJmL was calculated at 0.5 × 0.5 grid [44].
Terrestrial biodiversity (impact only)	Impacts via land use change, nitrogen deposition and climate change using the Mean Species Abundance (MSA) index as implemented in IMAGE-land (based on GLOBIO) [41, 45].

Table 2. *Scenarios.* CG and noCF refer to climate forcing (i.e. with climate impacts) and no climate forcing (without climate impacts), respectively.

Name	Technical reference	Forcing level in 2100 (W m^{-2})	Climate feedbacks	Based on climate pattern
Key scenarios				
Baseline	1. SSP2-6.0-noCF	6.0	Excluded	n.a.
Baseline with climate impacts	2. SSP2-6.0-CF	6.0	Included	IPSL
Mitigation	3. SSP2-26-noCF	2.6	Excluded	n.a.
Mitigation with climate impacts	4. SSP2-26-CF	2.6	Included	IPSL
Sensitivity scenarios				
Baseline with climate impacts (GDP excluded)	5. SSP2-6.0-CF-noGDP	6.0	Included (except GDP)	IPSL
Climate pattern sensitivity	6. SSP2-6.0-CF-H)	6.0	Included	Hadley

subsequently coupled to both MAGNET and TIMER. For MAGNET, also the impact on crop yields was included, while for TIMER, the impacts on cooling and heating degree days (CDD and HDD, respectively) and renewable energy potential on a grid-based level were coupled to the TIMER model, with implications for projected energy demand and supply [37–39]. In the impacts on cooling, the calculations take the affordability of cooling equipment into accounting (consistent with the so-called cooling gap, i.e. the fact the many people in low income countries cannot afford cooling technology [38, 40]. A simplified version of the mean species abundance (MSA) indicator has been used for biodiversity. While MSA is directly impacted by climate change, it is also impacted indirectly via land use changes induced by other climate impacts [41]. Finally, the climate change patterns also indirectly impact all other sectors via the changes introduced in energy and land use.

2.2. Scenario analysis

The baseline IMAGE 3.2 SSP2 scenario has a forcing level in 2100 near 6.0 W m^{-2}; hence this was used as the default forcing level [25]. The mitigation scenarios targeted a forcing level of 2.6 W m^{-2}, resulting in a 2100 global average temperature increase of around 1.7 °C. All scenarios are run with and without climate feedback (in the latter case, the climate is held constant from 2020 onwards) (table 2). The changes in global mean temperature are calculated internally in IMAGE using MAGICC—and subsequently patterns of temperature and precipitations are obtained by grid-based pattern scaling. The IPSL climate pattern is the default climate pattern used in IMAGE for precipitation and temperature, as it is one of the tier 1 climate patterns of ISIMIP2b [27]. The Hadley climate pattern is used in a sensitivity run (in both cases the pattern is combined with MAGICC based temperature outcomes). In addition, the analysis includes one baseline scenario that assumes all assessed climate impacts except those on GDP (SSP2-CF-IP-noGDP).

3. Results

3.1. Economic indicators and energy

In the baseline scenario GDP is assumed to grow in all regions in the absence of climate change. Climate change, however, is expected to have a significant

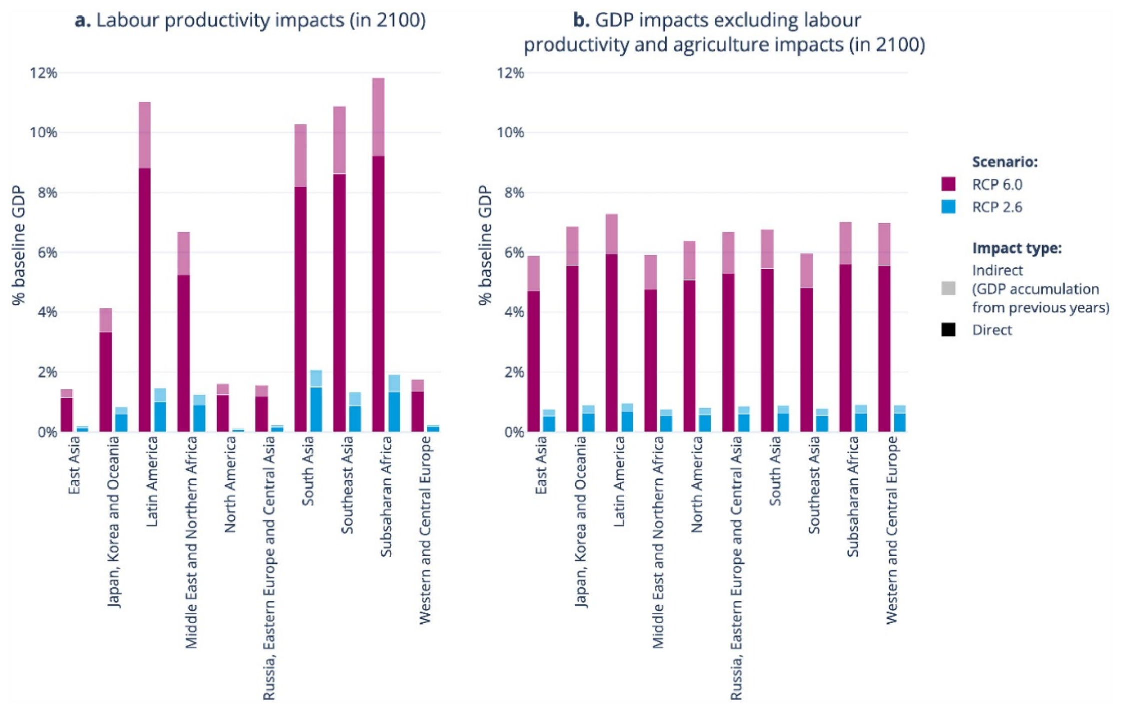

Figure 2. Climate impacts on labor productivity (left panel) and impacts on GDP including only impacts other than labor productivity and agriculture (right panel) for the baseline and 2.6 W m^{-2} scenario. GDP impacts are based on [35], scaled to the regional level using the data from Bosello and Decian [36] (see methodology section).

negative impact via reduction of labor productivity and the other factors included in the damage curves used (including flooding, energy systems, forestry, fisheries and transport) (see figure 2). The total GDP decline is up to around 15%–20% in some regions compared to the scenario without climate change. The labor productivity impacts are the most severe in tropical regions due to higher temperatures (and the non-linear impacts) and the higher prevalence of unsheltered (outdoor) labor (e.g. agriculture and construction). Other impacts on GDP growth are more balanced across regions. As expected, the climate impacts on labor productivity and GDP are much smaller in the mitigation scenario.

The energy system is directly affected by climate change through impacts on the buildings sector through both cooling and heating demands, as well as renewable energy sources. Figure 3 shows final energy use, which is in most regions significantly higher than today, driven by population and income dynamics (left side). In the mitigation scenario (RCP2.6), this increase is significantly reduced as result of efficiency improvements and further electrification of energy end-use in industry, buildings and transport. Including climate impacts (right panel) leads to a range of changes depending on the importance of cooling and heating in overall energy use. In temperate climates, final energy use decreases as reduced heating as result of climate change dominates. In warmer regions, overall final energy is projected to increase by around 5% due to increased cooling demand. The figure, however, also indicates that indirect impact of reduced GDP growth (see above, resulting from both reduced labor productivity and other economic impacts) could be significantly larger than the direct impacts. In that case, energy use is significantly reduced in all regions.

The changes are shown in figure 4 for the baseline and mitigation scenarios. Climate impacts both cooling/heating demand (see also figure 3) and the potential for several renewable energy options resulting in changes in primary energy demand. The impacts are quite mixed and depend on the region. For instance, in the baseline scenario, increased energy use for cooling in Sub-Saharan Africa leads mostly to more fossil fuel use; in the mitigation scenario, it leads to more renewable energy use.

3.2. Agriculture, food-related indicators and biodiversity

For agriculture, climate change can have significant impacts on crop yields, resulting in both negative (primarily in tropical regions) and positive impacts (in other regions) (figure 5). These impacts stem from changes in temperature, precipitation, and the fertilization effect of elevated CO_2 levels. The impacts are, in general, considerably larger in the baseline scenario than in the mitigation scenario. In tropical regions, the negative impacts of a hotter and drier climate often counterbalance the impacts of more CO_2 fertilization, except for tropical roots and tubers and sometimes soybeans that are expected to lead to

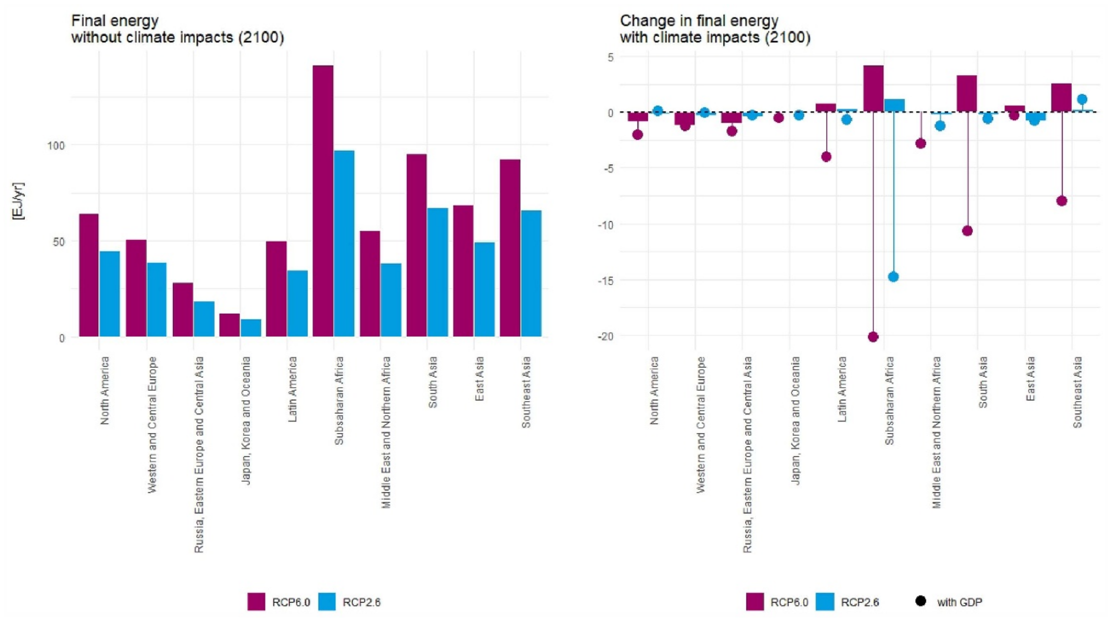

Figure 3. Final energy without climate impacts (left panel) and the changes resulting from including climate impacts for cooling, heating and renewable energy (right panel) (all in EJ yr^{-1}). The additional circles indicate the total change in final energy use when also the GDP impacts discussed in figure 2 are included.

higher yields under higher climate change circumstances. Conversely, most crops in temperate region experience positive impacts except for maize. It is critically important to realize that most crop models (like the one used in IMAGE) do not capture the yield impacts of extreme climate (especially droughts) well—and are likely to underestimate the impacts [46]. Moreover, IMAGE does automatically reallocate crops in the case of serious impact to cells with higher yields, so these results include one type of adaptation. Earlier, Rising and Devineni [47] showed that this could reduce climate impacts by half in the case of the USA under RCP8.5.

The changes in yield are expected to impact food prices and consumption (see also van Meijl *et al* [48]. Overall, most regions are expected to have an increase in food consumption in 2100 compared to 2015 (figure 5). This is a result of (economic) development and yield improvement. In many regions, the impact of development dominates over the impact of climate change. Still, in some regions the impact of climate change results in some different change in consumption (often an increase). The calculations also show the potential negative impacts of climate policy due to increased competition between food and land-based mitigation measures forest protection, afforestation, bioenergy.

The increase in afforestation in the mitigation scenarios is reflected in the results on the natural land share, as shown in figure 6. In terms of climate impacts, there is a small increase in nature area in the runs with climate change (mostly related to yield impacts). The runs including GDP impacts lead to a further increase in natural area, but this is a result of lower food consumption levels.

Biodiversity is affected directly by climate change—as well as indirectly via changes in other systems (including land use). Here, we assess biodiversity in the form of the MSA—a pressure-based measure of the state of biodiversity. In most regions, a decline in MSA of plants is projected for 2100 when compared to 2015, and this reduction is significantly exacerbated by climate change (figure 7).

3.3. Water stress and aridity

The impacts on the availability of water are directly influenced by changes in precipitation and evaporation—but also indirectly via changes in the food and energy systems. In several regions, the result is an increase in water scarcity. This is shown in figure 8 in the form of areas defined as arid regions based on the ratio of precipitation and potential evapotranspiration (aridity index, values smaller than 0.2 are defined as arid). As this index focuses on the biophysical components of water scarcity, there is no change without climate impacts. Including climate impacts, shows a significant increase in aridity in the baseline scenario (mostly as a result in increases of evapotranspiration). In the mitigation scenario, in some regions, however, a decrease of aridity is observed.

3.4. Overall results

We present a summary of the overall climate impacts in terms of a heat map (figure 9) for both the baseline

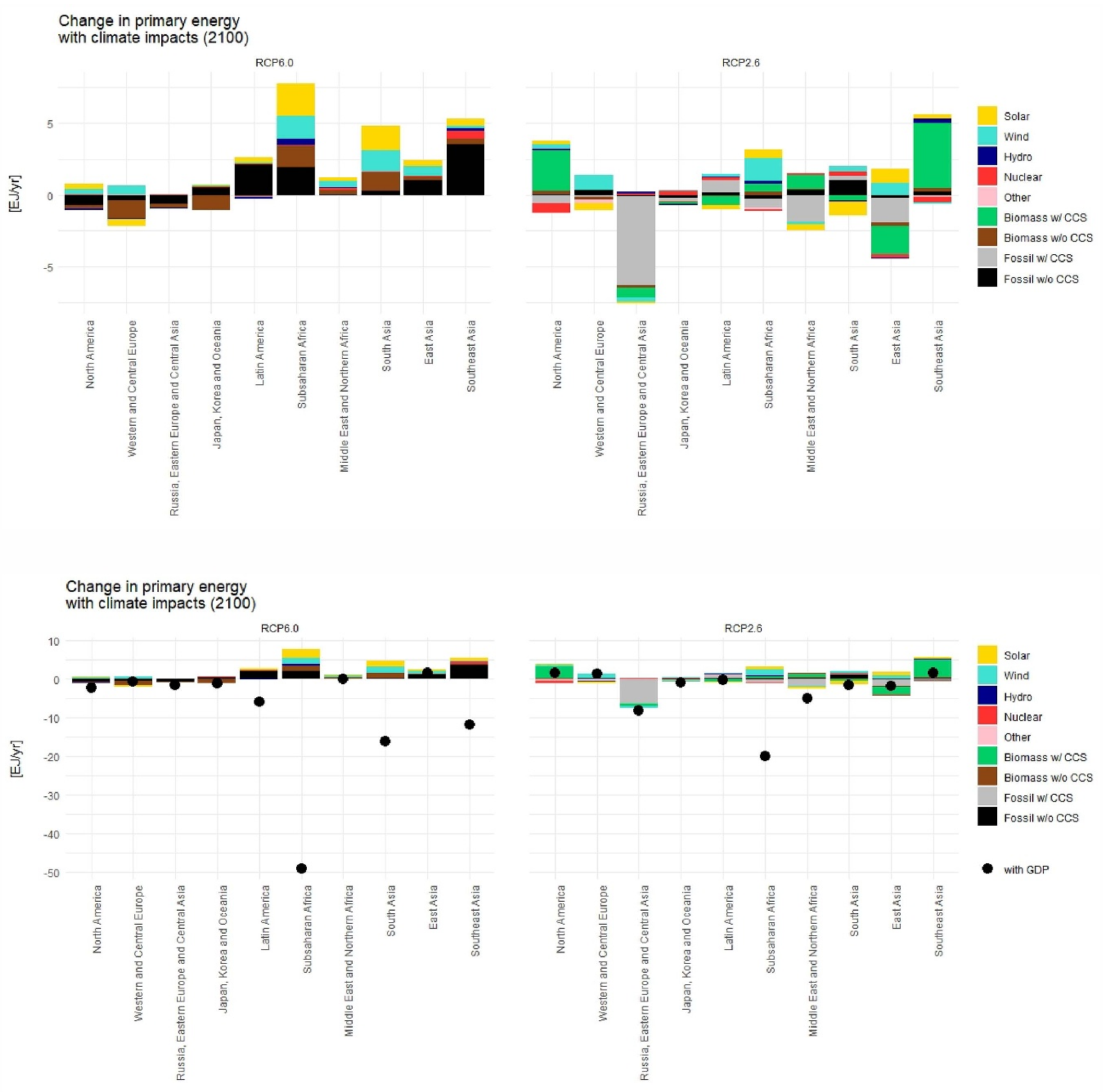

Figure 4. Changes in primary energy use in the RCP6 and RCP2.6 scenario without climate change impacts (top panel) and the changes in primary energy as result of including climate impact (bottom panel). The additional circles indicate the total changes if also the GDP impacts discussed in figure 2 are included.

scenario and mitigation scenario, highlighting the comparison between scenarios with and without climate change. Several patterns emerge from the analysis. Firstly, negative climate impacts generally exhibit correlation across different categories: temperate zones tend to experience lower impacts, while tropical regions emerge as climate impact hotspots. This trend is particularly evident in impact categories related to the economy, energy, crop yields, and, to some extent, food consumption. However, this correlation does not hold true for all impact categories. Impacts on water, biodiversity, and some of the energy indicators show less alignment with regional average temperatures. Another observation is that while the majority of the indicators are negative, there are also positive impacts, including reduced heating demand and, depending on the region, renewable energy production, crop yields, and water indicators.

The climate impacts are much less in the 2.6 W m^{-2} case. The impact distribution is largely in line with the baseline pattern. Globally, there are some notable differences in the climate impact categories. In the mitigation case, food consumption is 5% lower in the climate impact scenario, while net global food consumption is unaffected by climate impacts in the baseline. Climate impacts in the mitigation case also result in a small increase in natural land (0.26%), compared to a small decrease (−0.04%) in the baseline.

It should be noted that the colors on the desirable/undesirable impacts should be interpreted with care, as they are subject to the indirect impacts of GDP losses (undesirable). For instance, the reduction in final energy use is mostly an indirect result of the changes in GDP. In the supplementary information therefore also results without GDP impacts are presented.

Figure 5. Crop yields for selected crops (top row), and food demand in 2100 (bottom row), the left panels show the changes without climate change, and the right panels show the changes due to climate change. The additional circles in the bottom-right panel indicate the total changes if also the GDP impacts discussed in figure 2 are included.

4. Discussion and conclusions

This study's scenario assessment offers an initial understanding of the implications of integrating climate impacts within a process-based IAM. In the analysis, we focus on a selection of climate impacts that are included process-wise in IMAGE. Other impacts are included via a general function on GDP. In the figures, we have shown impacts with and without these GDP impacts included. This shows that in the current setup, the indirect impacts of climate change via general economic development form a very large, if not dominant, factor (summarized in annex 2). For instance, total greenhouse gas emissions would increase somewhat due to the physical climate impacts in the model, while they decrease if the economic impacts are also included.

Our results are broadly consistent with sectoral studies and the synthesis presented in IPCC AR6 [1]. For instance, the projected GDP losses are—by definition—similar to the meta-analyses of climate damages from 2017 [35], but also comparable to a recent multi-model assessment [14]. Likewise, our findings on increased cooling demand and renewable energy potential reflect patterns identified by other studies [19, 20, 39]. In agriculture, the IMAGE-based crop yield changes correspond to ISIMIP projections [49]. The novelty of this study lies in operationalizing these sectoral insights within a process-based IAM framework, enabling dynamic feedbacks between energy, land, and economic systems. Byers *et al* [39] looked at multiple impacts, but did not integrate them. Still, both studies and the IPCC report highlights that climate impacts will disproportionately affect tropical regions and exacerbate food, water, and

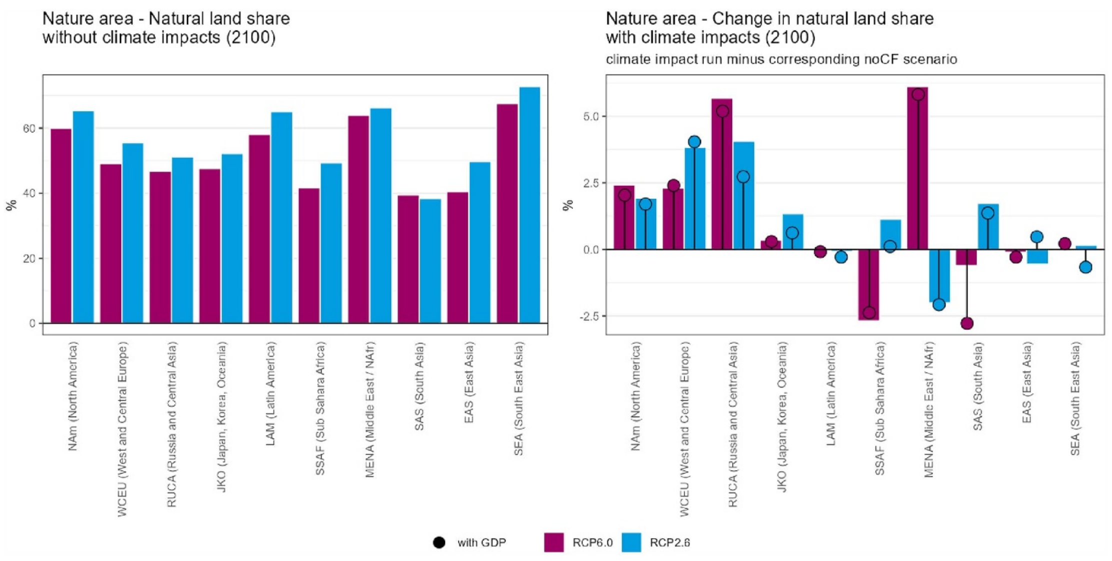

Figure 6. Natural land as share of total terrestrial area (%) in 2100 in RCP6 and RCP2.6 without climate impacts (left panel), and the changes due to including climate impacts (right). The additional circles indicate the total change if also the GDP impacts discussed in figure 2 are included.

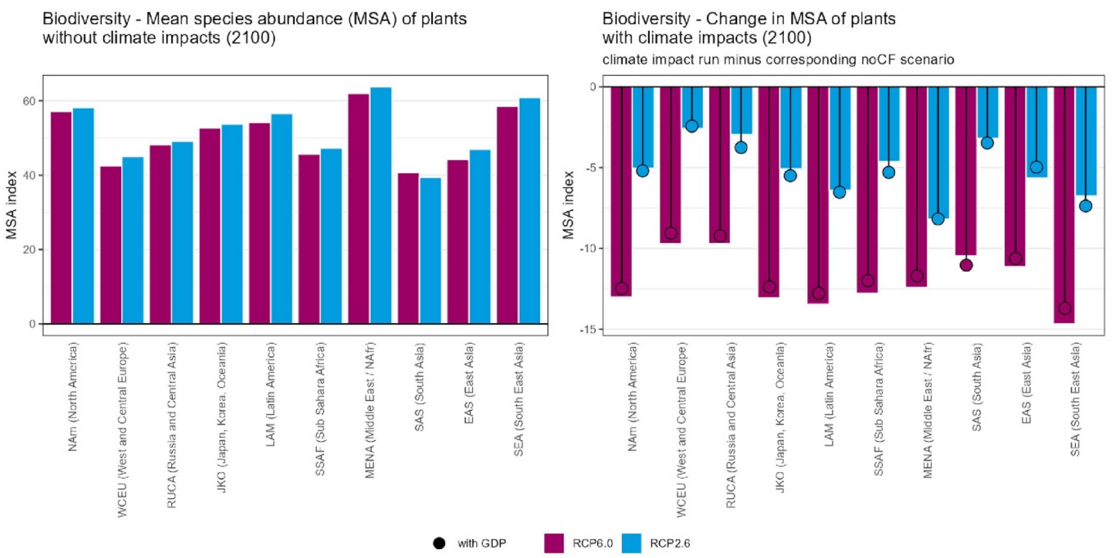

Figure 7. Mean species abundance (MSA) of plants in 2100 without climate change (left panel) and the impact of climate change (right panel). The additional circles indicate the total change if also the GDP impacts discussed in figure 2 are included.

biodiversity risks—patterns mirrored in our multi-impact hotspot analysis.

There are many uncertainties in this assessment. These include the socio-economic development assumed (e.g. population development and lifestyle), uncertainty in climate change (dominated by climate sensitivity, the pattern of climate change, and the carbon cycle), the uncertainty in each impact category, the indirect effects of these factors via the impact on socio-economic development, and the level of adaptation assumed. These uncertainties highlight the complexity of integrating climate impacts into IAMs and the need for careful consideration of various factors and assumptions. We discussed some of these uncertainties already in the main text. To look at the influence of the pattern of climate change, we have reproduced figure 6 for the baseline scenario using the Hadley pattern (RCP6.0). While the overall regional impact patterns and extent of impacts appear similar, there are notable differences in specific regions. For example, in Western Africa, the SSP2-CF-H scenario demonstrates an increase in food consumption compared to the baseline without climate feedback. Conversely, the SSP2-CF-IP scenario shows a relative decrease, highlighting the varying impacts projected by different climate change scenarios. Significantly, the Hadley pattern generally indicates a 'wetter' climate and predicts

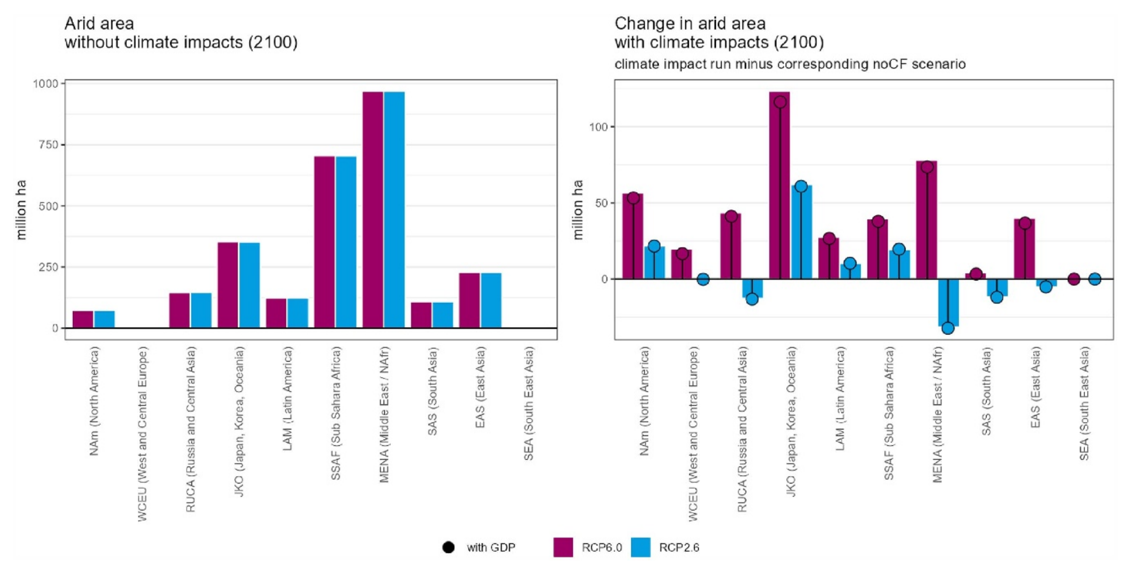

Figure 8. Total arid area (defined by the aridity index) under the baseline scenario and the mitigation scenario without climate impacts (left panel) and the difference in arid area due to climate impacts in 2100 (right panel). The additional circles indicate the total change if also the GDP impacts discussed in figure 2 are included.

a global reduction in drought (−6%) in the baseline scenario, while the warmer IPSL pattern projects a substantial increase (15%) in drought. Consequently, the Hadley-based scenario result in a lower increase in arid areas compared to IPSL. Additionally, the Hadley pattern shows a much larger decrease (−46%) in heating degree days (HDDs) than IPSL (−12%), suggesting that the regional distribution of reduced heating in the Hadley model favors more densely populated areas.

Overall, a number of limitations of the current study need to be acknowledged. First, of all, the study only looked at a **selected set of impacts.** The assessment provides a partial picture due to the selection of specific impacts. Notably, detailed description of impacts on health, severe weather events, sea-level rise, and the effects of flooding on infrastructure were not directly included in this analysis:. Sea-level rise, flooding and health impacts were still included via the overall economic damage function. Second, **adaptation is only partially incorporated.** Some forms of adaptation are clearly included by accounting for the use of air conditioning and decisions on agriculture locations. Other adaptation actions are lacking or not fully coupled (air conditioning is not directly coupled with labor productivity, although the assumptions on the cooling gap and the different impacts on labor in and outside buildings provide some consistency). Enhancing the representation of adaptation and linking it to specific SSP storylines would be valuable, but currently, the availability of such information is limited [7]. Third, the **GDP impacts are not explicitly coupled with sector-specific impacts.** Ideally, the sectoral damage curves would be integrated in an overarching economic model, enhancing consistency. While the current study demonstrates that it is possible to calculate a set of indicators based on a consistent description of future change, coupling the economic damage dynamically into the model would be the next step. The economic damages are now calculated separately. This may lead to some level of inconsistency. For instance, the damage curves also include impacts on energy systems (as a result of increased energy system costs) which could be not fully consistent with the results from IMAGE presented here (although calculations are done for the same scenario). It should be noted that there is no double counting. And finally, **it is critical to improve assessment of uncertainty.** As indicated above, there are important uncertainties that would impact this study. Here, we have only looked at some factors—given the explorative nature of the current study. However, in future work it is important to do this more systematically.

It can thus be concluded that this study forms one of the first attempts to examine multiple climate impacts simultaneously within scenarios of IAMs. By including a comprehensive range of impacts, it contributes to our understanding of the interconnectedness of climate change consequences.

This study provides a proof-of-concept of IAM scenarios that also include climate impacts. A noteworthy observation is the correlation of impacts in tropical regions. The study shows significant impacts, particularly in the context of the RCP 6.0 scenario. In the study, tropical regions emerge as

a. Impacts under RCP 6.0

		World	Russia	West-Europe	Japan	USA	China	Mid.-East	Brasil	India	Indones.	West-Africa
Economy	Labour productivity	-5%	-1%	-1%	-3%	-1%	-1%	-9%	-14%	-12%	-13%	-13%
	Total GDP	-25%	-3%	-7%	-5%	-8%	-6%	-21%	-24%	-35%	-28%	-24%
Energy	Non-biomass renewables	-16%	-5%	-6%	4%	2%	-1%	-42%	-9%	-18%	-9%	-11%
	Biomass renewables	-6%	-10%	-2%	-35%	-5%	-8%	-2%	5%	0%	-21%	-17%
	CDD (1)	35%	302	222	490	470	596	879	767	888	794	997
	HDD (1)	-26%	-1001	-462	-621	-571	-488	-452	-53	-91	0	-1
	Final energy total	-8%	-8%	-2%	-5%	-3%	0%	-7%	-9%	-11%	-7%	1%
Land	Crop yields	-3%	37%	1%	5%	4%	-3%	-12%	-11%	-9%	-8%	-16%
	Food consumption	0%	10%	-5%	2%	5%	1%	-2%	3%	0%	-4%	0%
	Aridity index (2)	15%	32	56	0	245	372	352	0	-55	0	253
	Natural land	0%	0%	1%	0%	0%	0%	1%	0%	-1%	0%	0%
	Biodiversity (plants)	-15%	-19%	-11%	-17%	-14%	-13%	-13%	-16%	-8%	-18%	-16%
	Total GHG (3)	-6%	-258	52	5	-196	342	340	-42	-993	-107	121

b. Impacts under RCP 2.6

		World	Russia	West-Europe	Japan	USA	China	Mid.-East	Brasil	India	Indones.	West-Africa
Economy	Labour productivity	-1%	0%	0%	0%	0%	0%	-1%	-3%	-3%	-3%	-3%
	Total GDP	-8%	2%	0%	3%	-1%	-2%	-7%	-8%	-5%	0%	-3%
Energy	Non-biomass renewables	-3%	-7%	1%	3%	-2%	4%	-3%	0%	1%	24%	2%
	Biomass renewables	-3%	-20%	9%	-6%	15%	-13%	-7%	-21%	-9%	-43%	0%
	CDD (1)	11%	93	68	151	145	183	270	236	273	244	306
	HDD (1)	-8%	-308	-142	-191	-176	-150	-139	-21	-32	0	-1
	Final energy total	-4%	-2%	0%	-4%	0%	-2%	-5%	-1%	-1%	1%	0%
Land	Crop yields	0%	13%	5%	2%	2%	2%	30%	-5%	-6%	-4%	-5%
	Food consumption	-5%	-1%	-5%	0%	-4%	-4%	-8%	-3%	-5%	40%	-13%
	Aridity index (2)	3%	0	2	0	33	88	63	0	-17	0	65
	Natural land	0%	0%	2%	0%	0%	0%	2%	-1%	0%	-2%	0%
	Biodiversity (plants)	-6%	-8%	-4%	-7%	-5%	-5%	-4%	-7%	-4%	-8%	-6%
	Total GHG (3)	9%	56	-161	10	-263	83	126	130	202	427	53

Figure 9. Climate impacts in the default RCP6.0 baseline case (upper) and 2.6 case (lower) and IPSL climate pattern in 2100. Impact indicators are listed vertically. Results are shown for the largest IMAGE regions within 10 aggregated world regions and the world (1st column). Regions are ordered based on yearly mean temperature (left = coldest, right = hottest). Values represent the difference in impact in the climate feedback (CF) cases compared to a no-CF case. Global data is represented as a relative change. Regional data is presented in relative change, except for cases where low or negative values in the no-CF case would lead to misleading relative values. This is the case for: cooling and heating degree days (CDD/HDD), aridity index and total GHG). Here absolute numbers are used. We have used color to show undesirable impacts (red), neutral or small impacts (white) and potentially desirable impacts (blue). It should be noted that the impact of GDP loss leads to a reduction of energy use in nearly all regions. *Note:* (1) Cooling/heating degree days per year, (2) Aridity index measured in 10^3 km^2, (3) Regional total GHG measured in GtCO2eq/year.

hotspots for climate-induced challenges, highlighting the vulnerability of economies and human well-being in these areas. This means that long-term adaptation plans and sustainable development strategies in these high-risk regions will be critical.

This study also emphasizes that further work is necessary to enhance the representation of climate impacts within integrated assessment scenarios. Key areas for improvement include a more comprehensive inclusion of adaptation strategies, ensuring consistency between overall economic impact projections and sector-specific effects, and exploring the interplay between climate impacts and adaptation efforts in the context of different socio-economic pathways. Moreover, it will be important to better represent climate extremes, which is still an important topic in impact research itself. In conclusion, while this study represents a valuable initial step, it highlights the pressing need for ongoing research and refinement. Future studies can provide more robust and nuanced assessments.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Supplemental information available at https://doi.org/10.1088/1748-9326/ae55b9/data1.

Conflict of interest

There is no conflict of interest for any of the authors.

Data access

The data presented in the study is available in IAMC template format. Detailed documentation of the IMAGE model can be found at the IMAGE website.

Ethical statement

There is no relevant ethical statement.

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